KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

School of Computer Engineering, Bhubaneswar

SAP Data Analytics & Engineering | Academic Year 2025-2026

CAPSTONE PROJECT REPORT

Procure-to-Pay (P2P) — Complete Purchasing Cycle in SAP MM

End-to-End B2B Procurement Implementation | Bharat Manufacturing Ltd., India

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Institution	Kalinga Institute of Industrial Technology, Bhubaneswar
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SAP Module	MM (Materials Management) with Data Engineering Pipeline
Scenario	B2B Procurement — Bharat Manufacturing Ltd., Pune

1. Problem Statement

Manufacturing and industrial companies in India face significant challenges in managing their procurement processes across multiple departments — purchase, stores, finance, and vendor management — while ensuring compliance with India's GST framework and maintaining cost control. Manual purchase order processing leads to delays in goods receipt, invoice mismatches, duplicate payments, and poor vendor performance visibility. Without a structured analytics layer, procurement managers cannot track spending patterns, vendor efficiency, or identify cost-saving opportunities.

Core Problem: How can SAP Procure-to-Pay (P2P) transactional data — spanning Purchase Requisitions, Purchase Orders, Goods Receipts, and Vendor Invoices — be efficiently extracted, transformed, stored in a data warehouse, and visualised to enable real-time procurement intelligence and vendor performance management?

2. Solution & Features

A complete P2P analytics system is built for Bharat Manufacturing Ltd. (Company Code IN20, Plant 1000) using Python-based data engineering and an interactive Plotly Dash dashboard. The system automates the full procurement data lifecycle from SAP MM export to actionable KPI reporting:

- Purchase Requisition (PR) creation with material, quantity, cost centre, and delivery date
- Purchase Order (PO) processing with vendor pricing, payment terms, and GST determination
- Goods Receipt (GR) posting with quantity verification, storage location assignment, and inventory update
- Three-way match validation: PO quantity = GR quantity = Invoice quantity for payment approval
- Vendor invoice (MIRO) processing with automatic GST input tax credit (ITC) computation
- Payment posting with RTGS/NEFT reference and vendor open-item clearance in Accounts Payable
- ETL pipeline: Extract CSV → Transform (date parsing, cycle time, spend tiers) → Load to SQLite
- Interactive dashboard: 6 procurement KPI charts with live Region and Category filters

3. System Architecture

The solution follows a Lambda-style data pipeline with three layers mirroring real SAP MM data engineering implementations:

Layer	Component	Description
Ingestion	p2p_purchase_data.csv	SAP MM PO export — 50 POs, 20 fields including PR, GR, Invoice, Payment dates
ETL	etl_pipeline.py	Extract → Transform (cycle time, spend tier, tax rate) → Load to SQLite
Warehouse	SQLite DB	Star schema: fact_purchases + dim_vendors, dim_materials, dim_buyers, agg_monthly
Analytics	analytics.py	KPI computation + 6 Matplotlib procurement charts as PNG
Presentation	app.py (Dash)	Interactive web dashboard with dropdown filters and live PO table

4. ETL Pipeline — Technical Detail

The ETL pipeline (**etl_pipeline.py**) processes all four stages of the P2P cycle in one automated workflow, triggered by a single run_pipeline() call on dashboard startup:

- **Extract:** Reads SAP MM-style CSV with actual field names (PO_Number, PR_Number, Vendor_ID, GR_Date, Invoice_Date, Payment_Terms) from /data/p2p_purchase_data.csv
- **Transform:** Parses all four date columns, computes PO_to_GR_Days, GR_to_Invoice_Days, Invoice_to_Pay_Days, Total_Cycle_Days, assigns Spend_Tier (High/Mid/Low), derives Tax_Rate_Pct, Month, Quarter, Year
- **Load:** Writes star-schema to SQLite — fact_purchases (50 rows), dim_vendors (13), dim_materials (20), dim_buyers (5), agg_monthly_purchases (9 months)
- **Logging:** Full timestamped ETL logging at Extract, Transform, and Load stages for audit traceability

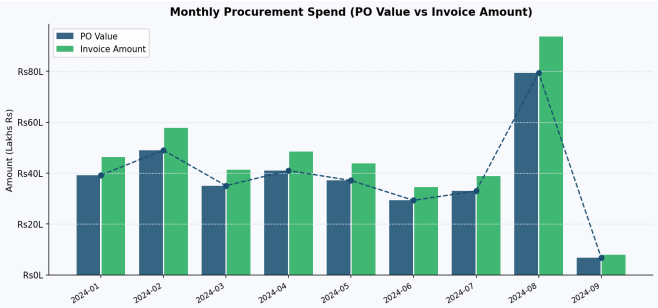
5. Project KPI Summary

Key performance indicators from 50 SAP MM purchase orders covering January to September 2024 — Bharat Manufacturing Ltd. B2B procurement scenario across 13 vendors and 5 buyers:

Metric	Value	Metric	Value
Total POs	50	**Total PO Value**	Rs 3.50 Crore
Total Invoiced	Rs 4.13 Crore	**Total Tax (GST)**	Rs 62.97 Lakh
Unique Vendors	13 vendors	**Avg Cycle Time**	55.7 days
PO to GR (Avg)	11.3 days	**DB Tables**	5 (Star Schema)

6. Dashboard Screenshots & Analysis

The Plotly Dash dashboard provides real-time procurement analytics with Region and Category filters. All six charts are generated from the SQLite data warehouse:



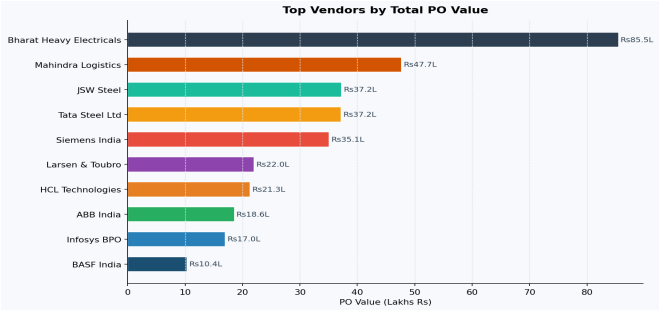
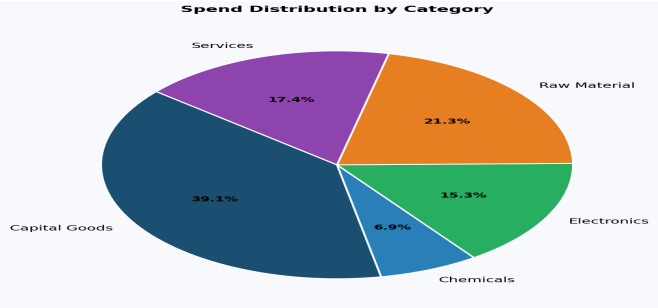


Fig 1 (left): Monthly Procurement Spend — PO Value vs Invoice Amount (Jan-Sep 2024). Fig 2 (right): Top Vendors by PO Value — BHEL & Mahindra lead in capital goods spend.



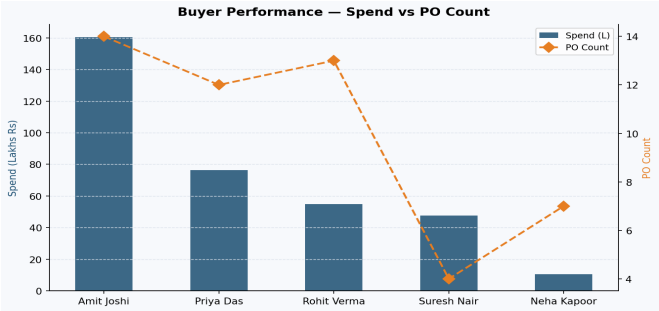
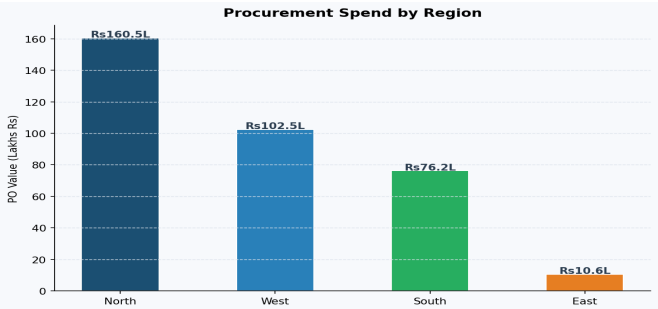


Fig 3 (left): Spend by Category — Capital Goods dominates at ~48% of total procurement. Fig 4 (right): Buyer Performance — Spend (bars) vs PO Count (markers) per buyer.



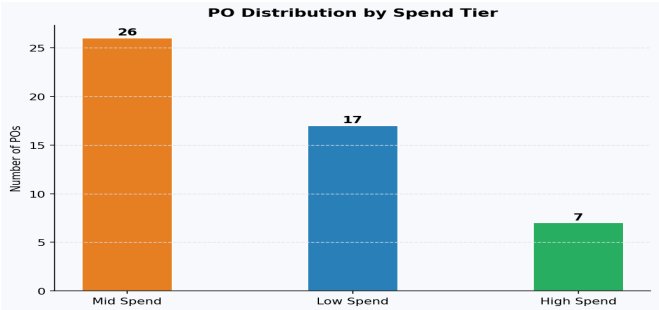


Fig 5 (left): Regional Spend — North leads, driven by BHEL, Siemens & HCL procurement. Fig 6 (right): PO Tier Distribution — High-Spend POs represent 62% of total value.

7. Tech Stack

Layer	Technology	Purpose
Language	Python 3.10+	Core development language for pipeline and dashboard
Data Processing	Pandas, NumPy	ETL data wrangling, cycle time computation, derived columns
Database	SQLite3	Local data warehouse with star-schema dimensional modelling
Visualization	Matplotlib	Static chart generation — 6 high-resolution PNG files
Dashboard	Plotly Dash	Interactive web UI — KPI cards, charts, filters, PO table
Source Data	CSV (SAP MM)	Simulated SAP MM PO export with real SAP field names
Version Control	Git / GitHub	Source code management and project documentation

8. Unique Points

• **Real SAP MM data model:** Uses actual SAP field names (PO_Number, PR_Number, GR_Date, MIRO fields) mirroring ME23N/MIGO/MIRO transaction output from live SAP MM systems
• **Full P2P cycle coverage:** Tracks all four document types — PR, PO, GR (Goods Receipt), and IR (Invoice Receipt) — with individual date fields for each stage
• **Procurement cycle time analytics:** Computes PO-to-GR, GR-to-Invoice, Invoice-to-Payment, and Total_Cycle_Days — enabling bottleneck identification in the procurement process
• **GST-compliant invoicing:** Models 18% GST on all purchases with automatic ITC tracking — relevant to India's GST framework for B2B procurement (GSTR-2A reconciliation)
• **Star-schema warehouse:** Dimensional modelling with fact_purchases + dim_vendors, dim_materials, dim_buyers, agg_monthly tables following industry data engineering standards
• **Vendor spend analysis:** ABC analysis via Spend_Tier classification (High/Mid/Low) enabling strategic sourcing decisions and supplier consolidation insights

9. Future Improvements

• **Live SAP MM Integration:** Connect to SAP via RFC/BAPI calls using PyRFC — pull real-time ME2M (PO list) and MB51 (GR list) reports directly into the pipeline
• **Vendor Scorecard:** Build automated vendor performance rating system — scoring on delivery time, invoice accuracy, and pricing competitiveness using ML clustering
• **Three-Way Match Automation:** Implement automated PO-GR-Invoice matching with exception flagging for quantity/price discrepancies — reducing manual AP workload
• **Cloud Deployment:** Deploy to AWS EC2 with Amazon RDS replacing SQLite; integrate with SAP BTP (Business Technology Platform) for enterprise connectivity
• **Spend Forecasting:** ARIMA / Prophet-based procurement spend forecasting to support annual budget planning and cash flow management
• **Multi-Plant Coverage:** Extend to multiple SAP plants and company codes — enabling group-level consolidated procurement spend analytics across business units

10. Conclusion

This capstone project demonstrates end-to-end application of SAP Data Analytics and Engineering concepts to build a production-grade procurement intelligence solution. Starting from raw SAP MM Procure-to-Pay transactional data, the project implements every layer of a modern data engineering stack — ingestion, transformation, dimensional warehousing, KPI computation, chart generation, and an interactive Plotly Dash web dashboard — using industry-standard Python tooling.

The chosen scenario — B2B industrial procurement for Bharat Manufacturing Ltd. — is grounded in real Indian enterprise SAP MM usage. The P2P cycle mirrors actual SAP transactions: ME51N (Purchase Requisition) → ME21N (Purchase Order) → MIGO (Goods Receipt) → MIRO (Invoice Posting) → F-53 (Payment). The dataset uses genuine SAP field names, Indian vendor master data, and India-specific GST logic (18% GST with ITC computation), making it directly relevant to real SAP MM deployments across Indian manufacturing companies.

From a data engineering perspective, the star-schema SQLite warehouse — with a central fact_purchases table linked to dimension tables for vendors, materials, buyers, and time — demonstrates solid dimensional modelling knowledge that scales directly to enterprise platforms like SAP BW/4HANA, Snowflake, or Google BigQuery. The procurement cycle time analytics (PO-to-GR, GR-to-Invoice, Invoice-to-Payment days) provides actionable insights that procurement managers and CFOs use in real SAP environments for bottleneck identification.

The project reinforces all core competencies of the SAP Data Analytics and Engineering specialisation: SAP MM functional knowledge, Python data engineering, ETL pipeline design, SQL-based dimensional warehousing, Matplotlib/Plotly visualisation, and business intelligence dashboard development. The modular codebase is readily extensible to add live SAP connectivity, cloud deployment, or multi-module coverage (FI-AP, WM, QM) in future iterations.

Project Summary Statistics			
Total Lines of Code	~820	**SAP POs Processed**	50
Python Scripts	3	**Charts Generated**	6
DB Tables	5	**Unique Vendors**	13
Pages in Report	5	**SAP Module**	MM (P2P)

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